CSA1718 AI

LAB PROBLEMS

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EXPERIMENT 11:

TIC Tac Toe Game

AIM

To develop a Tic Tac Toe game in Python where two players play alternately until one player wins or the game ends in a draw.

ALGORITHM

1. **Start.**
2. **Create an empty 3×3 game board.**
3. **Set the first player as 'X'.**
4. **Display the board.**
5. **Ask the current player to enter a position.**
6. **If the position is empty, place the player's symbol.**
7. **Check whether the player has won.**
8. **If a player wins, display the winner and stop.**
9. **Otherwise, switch to the other player.**
10. **Repeat until all 9 positions are filled.**
11. **If no player wins, declare the game as a draw.**
12. **Stop.**

PSEUDOCODE

START

Create empty board

Set current player = X

Repeat 9 times

Display board

Read player position

If position is empty

Place player's symbol

Else

Ask for another position

Check winning combinations

If player wins

Display winner

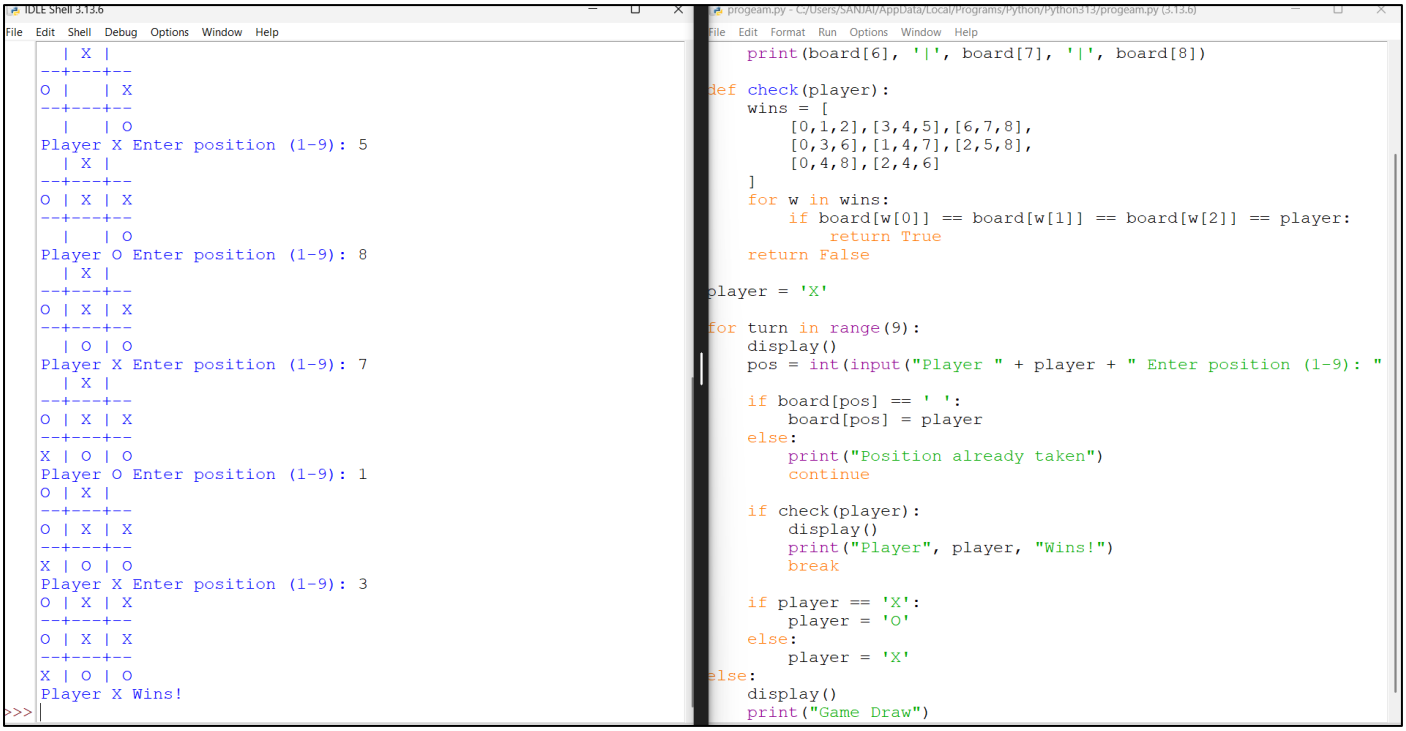
STOP

Change player

Display "Game Draw"

STOP

OUTPUT:



Result

The Tic Tac Toe game was successfully implemented in Python. The program allows two players to play alternately, correctly detects the winner, and declares a draw when no winning condition is achieved.

These are concise, lab-record-friendly Aim, Algorithm, Pseudocode, and Result sections suitable for practical examinations.

EXPERIMENT 12:

MIN MAX ALGO

AIM

To implement the Minimax Algorithm for the Tic Tac Toe game, enabling the computer to always choose the optimal move against the human player.

ALGORITHM

1. **Start.**
2. **Create a 3×3 Tic Tac Toe board.**
3. **Allow the human player to make a move.**
4. **Check if the game has reached a terminal state (win, lose, or draw).**
5. **If it is the computer's turn:**
    - **Generate all possible moves.**
    - **Apply the Minimax algorithm to each move.**
    - **If it is the computer's turn (MAX), choose the move with the highest score.**
    - **If it is the human's turn (MIN), choose the move with the lowest score.**
6. **Place the best move on the board.**
7. **Repeat until a player wins or the game ends in a draw.**
8. **Display the result.**
9. **Stop.**

PSEUDOCODE

```
START
Initialize board
FUNCTION Minimax(board, isMax)
  IF computer wins
    RETURN +1
  IF human wins
    RETURN -1
  IF draw
    RETURN 0
  IF isMax
```

best = -∞

FOR each empty cell

Place computer move

score = Minimax(board, False)

Remove move

best = max(best, score)

RETURN best

ELSE

best = +∞

FOR each empty cell

Place human move

score = Minimax(board, True)

Remove move

best = min(best, score)

RETURN best

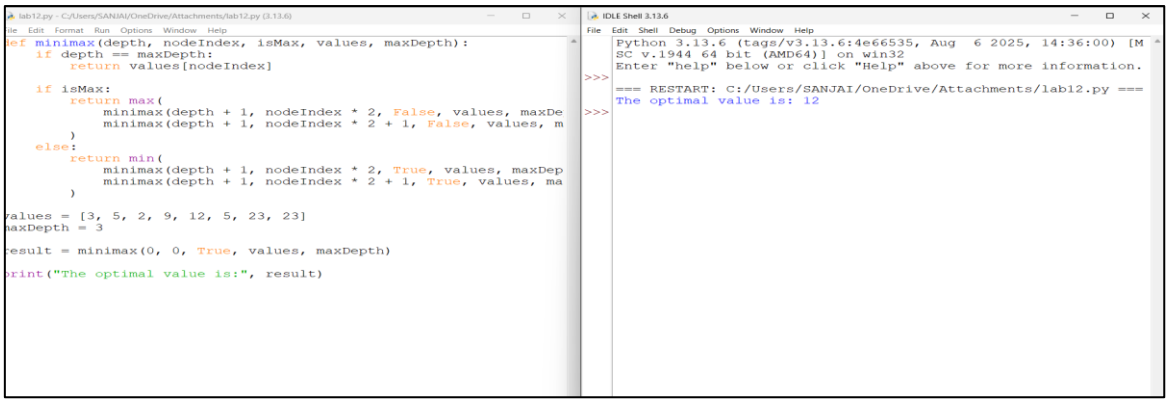
END FUNCTION

Computer selects move with highest Minimax score.

Display Winner or Draw.

STOP

OUTPUT



RESULT

The Tic Tac Toe game was successfully implemented using the Minimax Algorithm. The computer evaluates all possible future game states and always selects the optimal move, ensuring the best possible outcome (win or draw).

EXPERIMENT 13:

ALPHA BETA ALGORITHM

AIM

To implement the Alpha-Beta Pruning algorithm to optimize the Minimax search by eliminating branches that cannot affect the final decision and determine the optimal move in a game tree.

ALGORITHM

1. **Start at the root node with:**
    - **Alpha (α) = −∞**
    - **Beta (β) = +∞**
2. **If the current node is a terminal node, return its utility value.**
3. **If the current node is a MAX node:**
    - **Initialize the best value as −∞.**
    - **Evaluate each child recursively.**
    - **Update the best value.**
    - **Update α = max(α, best value).**
    - **If α ≥ β, stop exploring the remaining children (Beta Cutoff).**
4. **If the current node is a MIN node:**
    - **Initialize the best value as +∞.**
    - **Evaluate each child recursively.**
    - **Update the best value.**
    - **Update β = min(β, best value).**
    - **If β ≤ α, stop exploring the remaining children (Alpha Cutoff).**
5. **Return the best value obtained.**
6. **Continue until the root node returns the optimal move.**

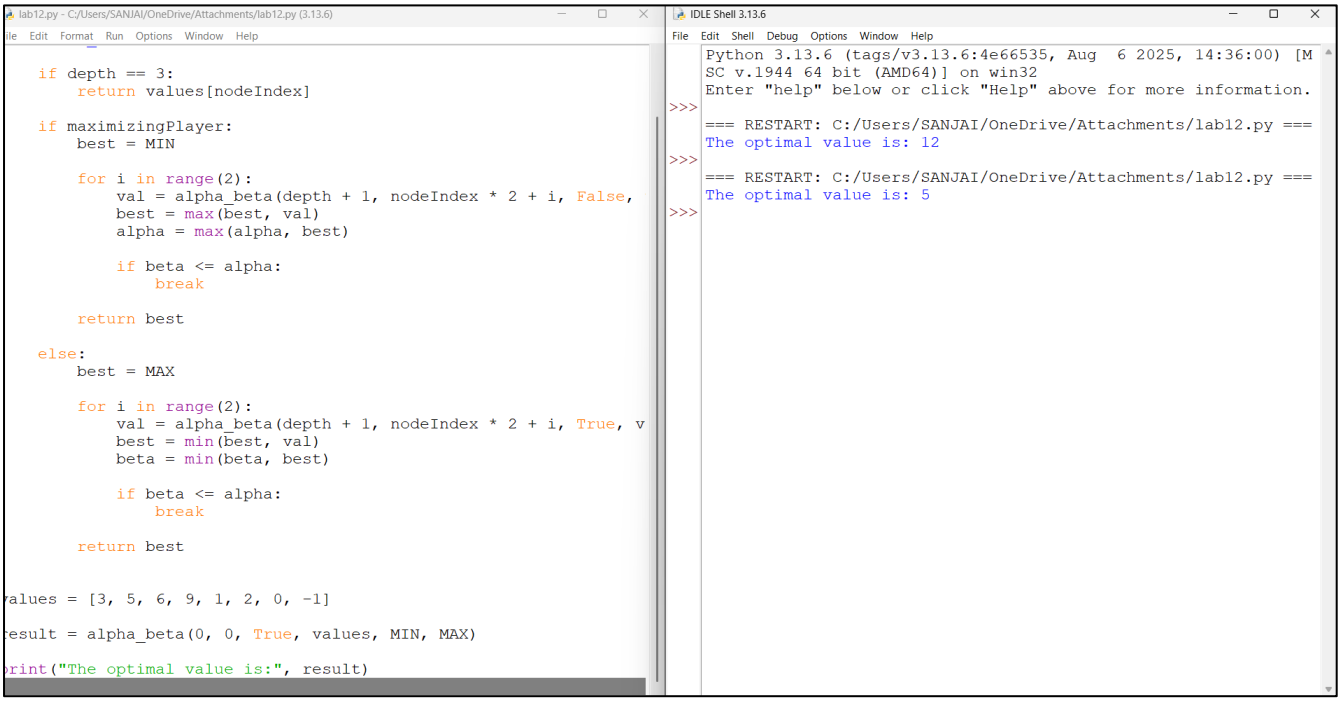
PSEUDOCODE

```
FUNCTION AlphaBeta(node, depth, alpha, beta, maximizingPlayer)
  IF node is terminal OR depth = 0 THEN
    RETURN node.value
  END IF
  IF maximizingPlayer THEN
    value ← -∞
```

```
    FOR each child of node DO
      value ← MAX(value,
              AlphaBeta(child, depth-1, alpha, beta, FALSE))
      alpha ← MAX(alpha, value)

      IF alpha ≥ beta THEN
        BREAK
      END IF
    END FOR

    RETURN value

  ELSE
    value ← +∞

    FOR each child of node DO
      value ← MIN(value,
              AlphaBeta(child, depth-1, alpha, beta, TRUE))
      beta ← MIN(beta, value)

      IF beta ≤ alpha THEN
        BREAK
      END IF
    END FOR

    RETURN value
  END IF

END FUNCTION
```

This is the standard Aim, Algorithm, and Pseudocode format commonly accepted for AI lab records.

OUTPUT:



```
lab12.py - C:/Users/SANJAI/OneDrive/Attachments/lab12.py (3.13.6)
File  Edit  Format  Run  Options  Window  Help

    if depth == 3:
        return values[nodeIndex]

    if maximizingPlayer:
        best = MIN

        for i in range(2):
            val = alpha_beta(depth + 1, nodeIndex * 2 + i, False,
            best = max(best, val)
            alpha = max(alpha, best)

            if beta <= alpha:
                break

        return best

    else:
        best = MAX

        for i in range(2):
            val = alpha_beta(depth + 1, nodeIndex * 2 + i, True, v
            best = min(best, val)
            beta = min(beta, best)

            if beta <= alpha:
                break

        return best


values = [3, 5, 6, 9, 1, 2, 0, -1]

result = alpha_beta(0, 0, True, values, MIN, MAX)

print("The optimal value is:", result)
```

```
IDLE Shell 3.13.6
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.6 (tags/v3.13.6:4e66535, Aug  6 2025, 14:36:00) [M
SC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/SANJAI/OneDrive/Attachments/lab12.py ===
The optimal value is: 12
>>>
=== RESTART: C:/Users/SANJAI/OneDrive/Attachments/lab12.py ===
The optimal value is: 5
>>>
```

Result:

The Alpha-Beta Pruning algorithm successfully determined the optimal move with a utility value of 5, while reducing the number of nodes evaluated through pruning compared to the standard Minimax algorithm.